Olist Order and Delivery Performance Analysis

01 Project Overview

This project analyzes Brazilian e-commerce order data from Olist to explore order trends, delivery performance, fulfillment delays, cancellations, and shipping costs. Using Python, PostgreSQL, and Power BI, the analysis transforms historical transaction data into actionable business insights to identify operational challenges and opportunities for improvement.

Dataset Source: Olist , containing historical orders from a Brazilian e-commerce platform connecting merchants with online marketplaces and logistic partners.

Business Problem

E-commerce businesses must balance fluctuating order demand, timely fulfillment, and affordable shipping while maintaining a reliable customer experience. Delivery delays, inefficient fulfillment processes, order cancellations, and high freight costs can create operational challenges.

Central Business Question:

How can Olist identify order groups and fulfillment stages that require attention, improve delivery reliability, and address shipping affordability through data-driven operational decisions?

What the project examines

- **Order trends:** Analyze monthly order volumes to identify demand patterns and periods of increased fulfillment workload.
- **Delivery performance:** Evaluate delivery timeliness and identify patterns associated with late deliveries.
- **Fulfillment efficiency:** Assess data inconsistencies (missing dates and conflicting event sequences) and examine delays across order approval, carrier pickup, and delivery stages.
- **Freight burden:** Compare freight charges relative to product values to identify order groups with higher shipping burdens.
- **Order Cancellations**: Examine cancellation patterns and the latest recorded fulfillment milestones among canceled orders.

Project Objective

Identify meaningful operational patterns, prioritize order groups and fulfillment processes requiring attention, and develop targeted business recommendations that can be evaluated through measurable performance indicators.

Project workflow: From Data Preparation to Business Insights



How Stakeholders Can Use the Results

The findings help operations teams identify delivery bottlenecks, prioritize high-risk order groups, and evaluate shipping affordability. These insights support targeted investigations and proposed operational improvements.

The effectiveness of these initiatives can be measured using late-delivery rates, late-order volumes, and freight cost per item, with consistent comparison periods and eligibility criteria.

Key Consideration: As the analysis is based on historical order-level records, the findings highlight associations rather than confirmed causes. Further investigation and pilot testing are needed to validate the proposed improvements.

02 Dataset Summary

Source: Brazilian E-Commerce Public Dataset by Olist

Scope: Two tables: orders and order_items

Table	Original size	Information included
orders	99,441 rows 8 columns	Order and customer IDs, order status, purchase and payment approval timestamps, carrier handoff, actual delivery, and estimated delivery dates.
order_items	112,650 rows 7 columns	Order and item IDs, product and seller IDs, shipping deadlines, product prices and freight charges.

Period and reporting scope

- **Full purchase period:** 4 September 2016 - 17 October 2018. The Python findings refer to the full dataset.
- **SQL analysis period:** January 2017 - August 2018, excluding sparse boundary months. Results from this period may differ from full-dataset figures.
- **Currency:** All product and freight values are recorded in Brazilian reais, shown as R$.

Table relationship and measurement

- **One order can contain several items:** The tables connect through order_id in a one-to-many relationship. An order can contain multiple items. To prevent double-counting in order-level analysis, item records are aggregated by order_id before joining with the orders table.

- **Order-Level Aggregation:** Product prices and freight charges are summed by order_id to calculate order-level values and ratios. Orders without matching item records are retained in overall order counts.

Data quality at a glance

- **Missing timestamps:** 160 approval dates, 1,783 carrier pickup dates and 2,965 customer delivery dates are missing from the orders table. The order_item table has no missing values.
- **Duplicates:** Neither table contains duplicate rows.

Key measurement rules

Term	Definition
Valid delivery	Orders marked delivered, with a recorded delivery date, delivery duration and a delivery timestamp on or after purchase.
Valid fulfillment	Delivered orders with purchase, approval, carrier pickup, and customer delivery timestamps present and in chronological order.
Late delivery	The actual delivery to the customer date falls after the estimated delivery date.
Recorded order value	Sum of product prices and freight charges per order. It does not represent recognised revenue or profit.

Limitations of the data

- **Analytical Scope:** The datasets support order, delivery, cancellation pattern, and shipping cost analysis but do not provide information on cancellation reasons, customer satisfaction, shipping distances, parcel sizes, or profitability.
- **Delivery Attribution:** Delivery timestamps are recorded at the order level rather than the individual item or seller level, making it impossible to determine which individual item, seller, or logistics partner caused a specific delay.
- **Historical Data:** The findings reflect historical order patterns and may not represent Olist's current operational performance.
- **Causal Limitations:** The analysis identifies patterns and potential fulfillment bottlenecks but cannot confirm the underlying causes without further investigation.

03 Data Cleaning Using Python

Python was used to clean, validate, and prepare the datasets for analysis. The process focused on identifying missing values, detecting data inconsistencies, and creating relevant features. Data quality issues were flagged rather than removed, preserving the original records and ensuring transparency. This approach supports more reliable and consistent analysis in PostgreSQL and Power BI.

Libraries: pandas, NumPy, Matplotlib

3A Inspect and validate

- **Data loading:** Load both CSV files and inspect row counts, data types and summary statistics to understand the structure of the data.

```
<class 'pandas.DataFrame'>
RangeIndex: 99441 entries, 0 to 99440
Data columns (total 8 columns):
 #   Column                         Non-Null Count  Dtype
---  ------                         --------------  -----
 0   order_id                       99441 non-null  str
 1   customer_id                    99441 non-null  str
 2   order_status                   99441 non-null  str
 3   order_purchase_timestamp       99441 non-null  str
 4   order_approved_at              99281 non-null  str
 5   order_delivered_carrier_date   97658 non-null  str
 6   order_delivered_customer_date  96476 non-null  str
 7   order_estimated_delivery_date  99441 non-null  str
```

```
<class 'pandas.DataFrame'>
RangeIndex: 112650 entries, 0 to 112649
Data columns (total 7 columns):
 #   Column               Non-Null Count   Dtype
---  ------               --------------   -----
 0   order_id             112650 non-null  str
 1   order_item_id        112650 non-null  int64
 2   product_id           112650 non-null  str
 3   seller_id            112650 non-null  str
 4   shipping_limit_date  112650 non-null  str
 5   price                112650 non-null  float64
 6   freight_value        112650 non-null  float64
```

- **Check identifiers and duplicates:** Confirm unique order IDs and unique order-item ID pairs. No duplicate rows were found.
- **Validate numeric values:** Check product prices and freight charges for invalid values. No non-positive prices or negative freight charges were found. Retain the 383 zero-freight items, as zero shipping cost does not necessarily indicate an error.

3B Prepare dates and review missing values

- **Date conversion:** Convert order timestamps and item shipping deadlines to datetime format to support date comparisons, monthly trends, and duration calculations.
- **Review missing dates:** Examine missing timestamps alongside order statuses to distinguish unfinished orders from potential data-quality issues.
- **Preserve evidence of exceptions:** Retain records with missing dates rather than removing them, as these may represent unfinished order stages or potential data-quality issues. This ensures important information remains available for further analysis.

3C Feature engineering

- **Delivery durations:** Calculate actual delivery time and the difference between actual and estimated delivery dates to measure delivery timeliness.
- **Delivery timing:** Classify orders as early, on time, late, or not delivered, while flagging inconsistencies between order status and delivery dates.
- **Purchase month:** Create a monthly grouping field to analyze order trends and changes in demand over time.

3D Database Integration

- **Data Type Mapping:** Specify the appropriate PostgreSQL data types for each column to ensure accurate and consistent data storage when loading the cleaned datasets into the database.
- **Data Transfer:** Connect Python to PostgreSQL and load the cleaned datasets into the database for further SQL analysis.

04 Exploratory Data Analysis (EDA)

Exploratory data analysis was conducted to understand overall order patterns, delivery performance, and data-quality issues. **The findings below describe the full dataset**, providing a baseline for deeper business analysis in PostgreSQL and highlighting areas requiring further investigation.

4A Order demand and delivery outcomes

- **Peak monthly demand:** November 2017 recorded the highest order volume (7,544 orders). To avoid misleading trends caused by incomplete boundary months, January 2017 to August 2018 was selected for monthly comparisons in SQL.
- **Order status:** Of the recorded orders, 96,478 were marked as delivered and 625 as canceled. However, order status alone does not guarantee that every timestamp needed for a duration calculation is present or consistent.
- **Delivery Timeliness:** Among 96,470 delivered orders with recorded delivery dates, 88,644 arrived early, 1,292 arrived on the estimated date, and 6,534 arrived late, resulting in a 6.77% late-delivery rate.

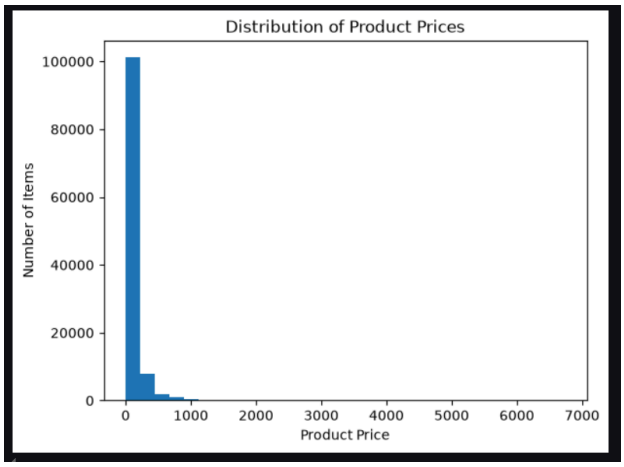
Delivery timing	Orders	Compared with the estimated date
Early	88,644	Before the promised date
On the promised date	1,292	On the promised date
Late	6,534	After the promised date

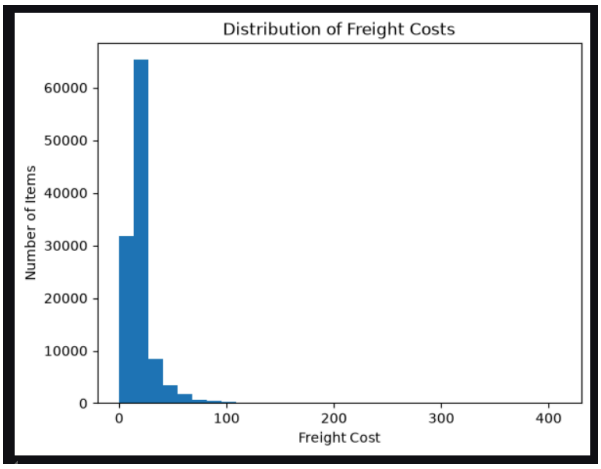
4B Data Quality Exceptions

- **Status and date conflicts:** Identified 8 delivered orders without delivery dates and 6 canceled orders with recorded delivery dates, highlighting discrepancies between order status and timestamps. These records need explicit treatment when reporting outcomes.
- **Invalid event sequences:** Found 1,359 orders with carrier handoff recorded before approval and 23 orders with delivery recorded before carrier handoff. These exceptions may overlap and should not be summed as unique orders. A record with an invalid stage sequence can remain visible in an exception summary while being excluded from a calculation that the sequence would distort.

4C Product Prices and Shipping Costs

- **Price and Freight Distribution:** Both distributions were right-skewed, with most items having lower values and a smaller number having much higher values. The median item price was R$74.99, while the median freight charge was R$16.26. Medians provide a more representative measure of typical costs when unusually high values affect the average.





- **Price and Freight Relationship:** A moderate positive correlation (r = 0.41) indicates that higher-priced items tended to have higher freight charges. However, this relationship does not establish causation.
- **Freight-to-Product Ratio:** Product prices and freight charges are aggregated by order before calculating the ratio to assess shipping costs relative to order value during SQL analysis.

4D Key Takeaways for Further Analysis

- **Delivery Reliability and Impact:** Use late-delivery rates to compare performance across order groups and late-order counts to identify where delays have the greatest impact. Apply consistent time periods and eligibility criteria for fair comparisons.
- **Shipping Cost Analysis:** Aggregate item-level prices and freight charges by order to assess shipping costs relative to product value. Distinguish between average order-level freight ratios and the overall freight-to-product ratio for accurate comparisons.

05 Business Analysis Using PostgreSQL

PostgreSQL was used to conduct deeper business analysis using the cleaned datasets from Python. The analysis explores order trends, delivery reliability, fulfillment delays, cancellations, and shipping costs to identify operational patterns and areas for improvement.

The following business questions build on the exploratory findings, providing **actionable insights and evidence-based recommendations** to support operational decision-making.

Q1. Which months experienced both higher-than-average order volume and above-average late-delivery rates?

	Purchase month text	Total orders bigint	Delivered orders bigint	Late orders bigint	Late rate (%) numeric
1	2017-11	7544	7288	904	12.40
2	2017-12	5673	5513	411	7.46
3	2018-02	6728	6555	926	14.13
4	2018-03	7211	7003	1328	18.96

- Four months had above-average orders and exceeded the average late rate of **6.79%**.
- March 2018 was the worst: **1,328 late orders (18.96%), 1 in 5 delivered orders**, followed by February (**14.13%**).
- A Correios strike began on 12 March 2018, providing possible external context for some delays. However, these tables do not identify the carrier used for each order, so the strike's contribution cannot be confirmed.
- November 2017 had the most orders, but a lower late rate than February and March. Volume alone does not explain delays.

Action: Compare fulfillment stages in March, February, and November to locate delays.

Q2. Do orders with multiple items take longer to arrive or have a higher late delivery rate than single-item orders?

	Item group text	Delivered orders bigint	Median delivery days numeric	90th percentile delivery days numeric	Late rate (%) numeric
1	Multiple items	9605	10.00	21.00	5.30
2	Single item	86598	10.00	23.00	6.95

- Both groups had a **10-day median delivery time**, but multiple-item orders had a lower late rate: **5.30% vs 6.95% (single-item orders)**.
- About 90% arrived within **21 days** for multiple-item orders and **23 days** for single-item orders.

Action: Investigate single-item orders first; they represented about **90%** of valid delivered orders. Stakeholders can compare their pickup and transit times with multiple-item orders to identify where the difference occurs.

Delivery Priority Groups

Q3. How does delivery performance differ by product value?

	Product value band text	Delivered orders bigint	Late orders bigint	Late rate (%) numeric	Median delivery days numeric	90th percentile delivery days numeric
1	Below R$50	28708	1731	6.03	9.00	22.00
2	R$50 to below R$150	43974	3000	6.82	10.00	23.00
3	R$150+	23521	1800	7.65	11.00	25.00

- As product value increased, late rates rose from **6.03% to 7.65%**, and median delivery time increased from **9 to 11 days**.
- Around **90% of R$150+ orders arrived within 25 days**, compared with **22 days** for orders below R$50.
- The middle band had the most late orders (**3,000**) and delivered orders (**43,974**); R$150+ orders had the highest late rate.
- Orders below R$50 arrived faster and have a lower late rate despite a higher freight ratio (based on dashboard)**.**
- R$150+ orders have a lower relative freight burden, but weaker delivery performance. Larger or heavier products are one possible explanation, but weight and dimensions are outside this analysis, so this cannot be tested here.

Action: Target the middle band for volume reduction and R$150+ orders for delivery-risk investigation.

Q4. Which product value and item-count groups had higher late rates?

	Product value band text	Item group text	Delivered orders bigint	Late orders bigint	Late rate (%) numeric
1	Below R$50	Multiple items	1150	60	5.22
2	Below R$50	Single item	27558	1671	6.06
3	R$50 to below R$150	Multiple items	4213	236	5.60
4	R$50 to below R$150	Single item	39761	2764	6.95
5	R$150+	Multiple items	4242	213	5.02
6	R$150+	Single item	19279	1587	8.23

- Single-item orders had higher late rates in every value band.
- Single-item R$150+ orders had the highest late rate (**8.23%; 1,587 late orders**), versus **5.02%** for multiple-item orders.
- Among single-item orders, late rates increased with product value. Multiple-item rates stayed around **5–6%**, showing no similar upward pattern.
- Single-item middle-value orders had the most late deliveries (**2,764**).

Action: Target high-value single-item orders for late-delivery risk and middle-value single-item orders for volume reduction.

Fulfillment delays

Q5. Which fulfillment stages take longer for late deliveries, and does this pattern differ by product value?

	Product value band text	Delivery group text	Delivered orders bigint	Avg approval days numeric	Avg pickup days numeric	Avg transit days numeric
1	Below R$50	By promised date	26586	0.41	2.44	7.41
2	Below R$50	Late	1725	0.59	4.59	26.86
3	R$50 to below R$150	By promised date	40324	0.38	2.55	8.16
4	R$50 to below R$150	Late	2988	0.46	5.34	27.99
5	R$150+	By promised date	21403	0.40	2.98	8.48
6	R$150+	Late	1793	0.49	6.82	28.69

- **Transit showed the largest difference across every value band.** 27-29 days for late orders, versus 7-8.5 days for delivery by the promised date.
- Pickup/carrier-handoffs were also slower among late orders, especially for R$150+ orders (**6.82 vs 2.98 days**). Approval differences were small.
- This strengthens the earlier finding that higher-value orders have weaker delivery performance: their late orders had the longest pickup and transit times.
- R$150+ orders have the highest late rate, and their stage breakdown points to pickup and transit as the main areas to investigate.
- **Fulfillment Time by Stage** chart already shows transit as the longest stage. This query adds meaning by showing that **late orders spend much longer in transit and also wait longer for pickup/carrier-handoffs**.

Action: Prioritise pickup and transit (especially for R$150+); escalate orders nearing their promised date.

Q6. How does the time from approval to carrier pickup relate to late rates?

	Days to ship text	Product value band text	Delivered orders bigint	% of shipping group numeric	Late rate (%) numeric
1	Up to 1 day	Below R$50	8546	30.75	3.74
2	Up to 1 day	R$50 to below R$150	12989	46.73	3.77
3	Up to 1 day	R$150+	6260	22.52	4.22
4	Over 1 to 3 days	Below R$50	11276	30.50	5.13
5	Over 1 to 3 days	R$50 to below R$150	16970	45.91	5.77
6	Over 1 to 3 days	R$150+	8721	23.59	6.28
7	Over 3 days	Below R$50	8489	28.24	9.74
8	Over 3 days	R$50 to below R$150	13353	44.43	11.37
9	Over 3 days	R$150+	8215	27.33	11.94

- **Longer pickup times = higher late rates (in every product value band).** Pickup taking over **three days** was linked to late rates **2.6–3 times higher** than pickup within one day.
- Within the over-three-day group, R$150+ orders had the highest late rate (**11.94%**); the middle band had the largest slow-pickup group (**13,353 orders; 44.43%**).
- This query adds a practical warning sign: **pickup taking more than three days is associated with substantially higher late delivery rates**, across all value bands.
- The previous query identified longer pickup times among late orders. This query strengthens that story by showing a consistent increase in late rates as pickup time increases.

Action: Pilot pickup monitoring for middle-value orders waiting over three days (largest group).

Shipping Affordability

Q7. Where did freight charges exceed product value?

	Product value band text	Orders with items bigint	High-freight orders bigint	High-freight rate (%) numeric
1	Below R$50	28708	2911	10.14
2	R$50 to below R$150	43974	152	0.35
3	R$150+	23521	6	0.03

- Freight exceeded product value for **10.14%** of orders below R$50, versus **0.35%** in the middle band and **0.03%** for R$150+.
- Low-value orders accounted for **94.9% of all high-freight order**s.
- The **Freight-to-Product Ratio** chart shows an average ratio of **57.1%** for orders below **R$50**. This query adds meaning by showing how often the burden becomes particularly high: **10.14% of orders in that band had a ratio above 100%**.

Action: Test same-seller bundles or combined shipping, where feasible.

Shipping charges and cancellations

Q8. How did freight charges relate to delivery time?

	Product value band text	Freight value band text	Delivered orders bigint	Median delivery days numeric
1	Below R$50	Below R$20	24990	9.00
2	Below R$50	R$20 to below R$40	3299	14.00
3	Below R$50	R$40+	419	14.00
4	R$50 to below R$150	Below R$20	29035	9.00
5	R$50 to below R$150	R$20 to below R$40	11916	12.00
6	R$50 to below R$150	R$40+	3023	13.00
7	R$150+	Below R$20	9226	9.00
8	R$150+	R$20 to below R$40	8346	12.00
9	R$150+	R$40+	5949	13.00

- **Higher freight charges = longer delivery times (every product value band)**
  Median delivery time was **9 days** with freight below R$20, versus **12–14 days** in higher freight bands.
- **Low-value orders showed the biggest gap:** median delivery time increased from **9 to 14 days** when freight was R$20 or more.
- **All three product value bands had the same 9-day median when freight was below R$20.** This suggests product value alone does not explain the delivery-time differences.
- The below-R$50, R$40+ group contained only **419 orders (smallest group).**
- This query adds another perspective: **orders with higher actual freight charges also tended to take longer to arrive**, even within the same product value band.
- It also explains variation hidden by the dashboard's **10-day Median Delivery Days** KPI: some groups had a 9-day median, while others reached 14 days.

Action: Review low-value orders with freight of R$20+. Higher charges did not necessarily mean faster delivery.

Q9. Which product value band had the highest cancellation rate without recorded delivery, measured against all eligible orders in that band?

	Product value band text	Orders with items bigint	Cancelled orders with no delivery recorded bigint	Cancellation rate (%) numeric
1	R$150+	24160	142	0.59
2	Below R$50	29303	129	0.44
3	R$50 to below R$150	44890	176	0.39

- R$150+ orders had the highest cancellation rate (**0.59%**); the middle band had the lowest (**0.39%**) but the most cancellations (**176**).
- Cancellation rates were **below 1% in every band**.
- The **Overall Cancellation Rate (0.59%)** card summarises cancellations. This query adds detail by identifying **which product value groups have a higher cancellation rate**.
- It also complements **Canceled Orders by Fulfillment Milestone**: the dashboard shows the latest milestone recorded, while this query shows the product value groups involved.

Action: Investigate high-value orders for cancellation rate and target the middle band to reduce cancellation volume.

Business priorities

Q10. Which recorded fulfillment milestones dominated cancellations?

	Latest recorded milestone text	Canceled orders bigint	Cancelled without items bigint	Product_value (R$) numeric
1	Approval recorded	394	16	85295.35
2	Carrier pickup recorded	69	0	6857.76
3	Delivery recorded	1	0	29.99
4	No approval recorded	116	116	[null]

- Approval without recorded carrier pickup or delivery accounted for **394 out of 580 cancellations (67.9%)** and **92.5%** of recorded canceled product value.
- **132 orders lacked item records**, so their value was unavailable. Another **69 had pickup recorded**, and **one had delivery recorded**.
- This query directly supports **Canceled Orders by Fulfillment Milestone** and explains the **R$92.18K Canceled Product Value** card: most of that recorded value comes from the approval-recorded group.

Action: Review approved orders canceled without recorded carrier pickup. Check stock availability and seller readiness after approval; review shipment handling and the conflicting delivery record.

Q11. Which groups should stakeholders prioritise? (above-average freight-to-product ratio or above-average late delivery rate)

	Product value band text	Item group text	Delivered orders bigint	Late orders bigint	Average freight ratio (%) numeric	Late rate (%) numeric
1	Below R$50	Multiple items	1150	60	89.95	5.22
2	Below R$50	Single item	27558	1671	55.62	6.06
3	R$50 to below R$150	Multiple items	4213	236	38.27	5.60
4	R$50 to below R$150	Single item	39761	2764	22.01	6.95
5	R$150+	Single item	19279	1587	10.95	8.23

- Multiple-item orders below R$50 had the highest average freight ratio (**89.95%**) but a relatively low late rate (**5.22%**).
- Single-item R$150+ orders had the highest late rate (**8.23%**).
- Single-item middle-value orders had the most late deliveries (**2,764**).
- Multiple-item R$150+ orders exceeded neither overall average.
- This query brings together **Average Items per Order**, **Freight-to-Product Ratio**, and **Late Delivery Rate**. It shows that shipping affordability and delivery reliability require different priorities:

Action: Test same-seller bundles or combined shipping where operationally feasible for low-value multiple-item orders. Pilot carrier pickup monitoring for the priority single-item groups.

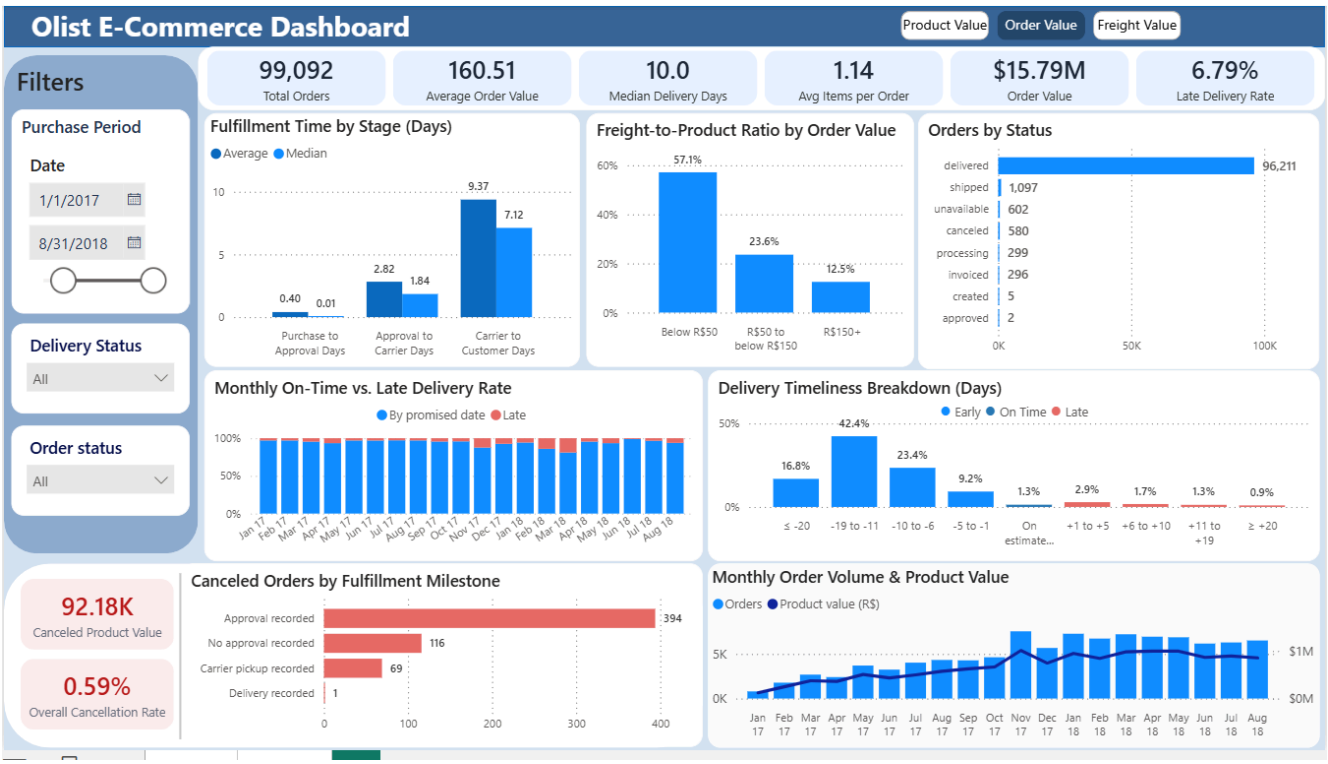
Key Takeaways

Target group	Problem	Action	Measure success
Single-item R$150+	Highest late-delivery rate	- Investigate pickup and transit delays - Pilot alerts for approaching their promised delivery date	A lower late-delivery rate
Single-item middle-value (R$50 to below R$150)	Most number of late deliveries	- Pilot pickup monitoring - Flag orders awaiting pickup more than 3 days after payment approval	Fewer late orders and a lower late-delivery rate, accounting for changes in order volume
Multiple-item below R$50	Highest relative freight burden	- Test same-seller bundles or combined shipping, where feasible	Lower freight charges per item without increasing the late-delivery rate

06 Interactive Dashboard Using Power BI

Power BI was used to develop an interactive dashboard that brings together key operational metrics and insights from the analysis. The dashboard provides a comprehensive overview of **order volume, recorded order value, delivery performance, fulfillment efficiency, cancellations, and shipping costs**.

Through interactive filters and visualizations, stakeholders can **monitor key performance indicators (KPIs), track monthly trends, identify potential fulfillment bottlenecks, and compare shipping costs across order value groups**. These insights help highlight operational challenges, prioritize areas for investigation, and support data-driven decision-making.



07 Business Recommendations

1. Launch a “Delivery Rescue” Program

- **Key Finding:** Single-item orders worth R$150+ recorded the highest late-delivery rate (8.23%).
- **Proposed Action:** Pilot proactive shipment monitoring for this order group. Flag orders approaching their estimated delivery date, investigate stalled shipments, escalate potential delays to logistics partners, and provide customers with timely updates.
- **Success Metrics:** Late-delivery rate and number of late orders within the targeted group.

2. Improve Carrier Handoff and Transit Efficiency

- **Key Finding:** Orders with approval-to-carrier handoffs exceeding three days showed higher late-delivery rates across product value bands. Late deliveries also had longer average transit times (27.89 days versus 8.01 days for timely deliveries).

- **Proposed Action:** Pilot seller follow-up reminders and improved pickup coordination for prolonged handoffs. Separately investigate unusually long carrier transit times to identify potential logistics issues before introducing targeted interventions.
- **Success Metrics:** Average handoff duration, average transit duration, and late-delivery rate.

3. Pilot a “Bundle and Save on Shipping” Program

- **Key Finding:** Among orders below R$50, 10.14% had freight charges exceeding product value, indicating a higher relative shipping burden.
- **Proposed Action:** Test same-seller product bundles or capped shipping discounts for low-value purchases. Verify that combined shipping reduces freight costs and assess profitability before expanding the program.
- **Success Metrics:** Freight cost per item and the percentage of orders with freight exceeding product value. Assess profit margin using additional product-cost, shipping-cost, and discount data before expanding the pilot.

4. Investigate Cancellations After Approval

- **Key Finding:** A total of 394 canceled orders had approval recorded but no subsequent fulfillment milestone.
- **Proposed Action:** Review cancellation reasons to identify recurring issues. If inventory shortages are common, pilot stock verification before approval. If seller dispatch delays are identified, test automated follow-up reminders.
- **Success Metrics:** Cancellation rate, number of approved orders subsequently canceled, and canceled product value.

5. Strengthen Peak-Period Fulfillment Planning

- **Key Finding:** Historical analysis identified periods of high order volume and elevated late-delivery rates, highlighting opportunities to investigate fulfillment capacity.
- **Proposed Action:** Use historical demand and delivery patterns to anticipate busy periods. Pilot additional seller dispatch support and carrier pickup capacity, prioritizing order groups contributing the highest number of late deliveries.
- **Success Metrics:** Late-order count, late-delivery rate, and average fulfillment duration during comparable peak periods.